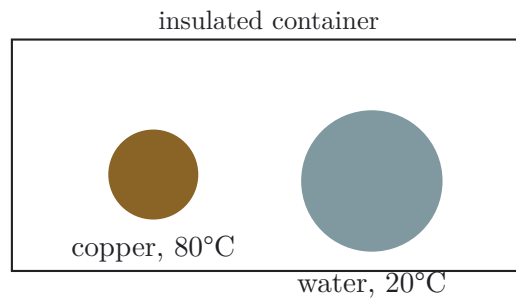


How Does It Know Where to Stop?

Name: _____

Two temperatures, one insulated box, and a stopping point that turns out to be a matter of overwhelming likelihood rather than any law pushing it there.



The question. When two objects at different temperatures are allowed to exchange energy, how does the system “decide” where to stop?

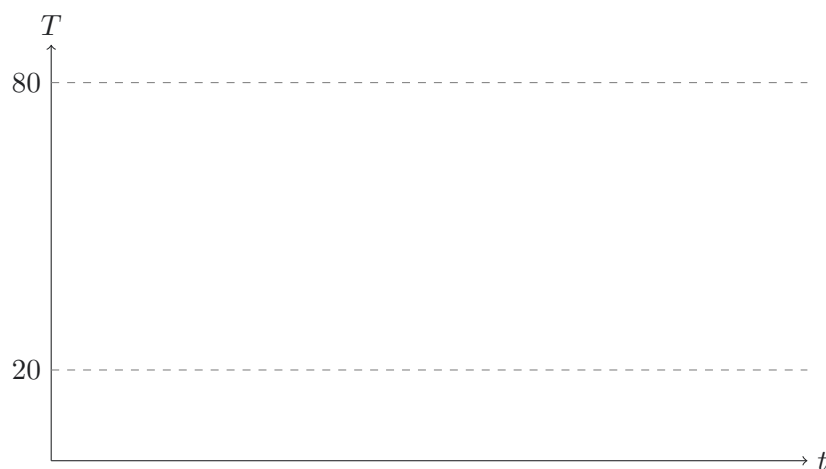
Working rule: Equal temperature is not assumed to be the definition of equilibrium. No formula for the equilibrium temperature is supplied early, and nothing is assumed to push energy toward balance.

Building on: This sheet reuses Ω , the plain count of hidden arrangements, from *How Many Ways Can It Look the Same?* without re-deriving it.

1. Sketch it before you calculate it

temperature against time

Sketch your prediction for each object on the same axes.



- Must the two temperatures meet?
- Must they meet exactly halfway?
- Could the lines cross?
- Why does the transfer stop when it does, rather than earlier or later?

2. What “equilibrium” actually means

an operational definition

Complete the definition without using the word “temperature”.

Thermal equilibrium: the macrostate in which _____

Why is “equal temperature” a symptom of equilibrium rather than a definition of it?

3. The first surprise: not necessarily halfway

two thought experiments

Two thought experiments, changing only one detail between them.

Case A Identical cups of water, one at 80°C, one at 20°C. Predicted equilibrium: _____

Case B A tiny hot copper ball at 80°C dropped into a large beaker of water at 20°C. Is 50°C still plausible? _____

What single new idea does case B force you to introduce, and why?

4. A model arrives

mass and specific heat capacity

Heat lost by the hotter object equals heat gained by the cooler one. Total heat capacity C splits into mass m (extensive) and specific heat capacity c (intensive, a property of the material alone).

$$C = mc \qquad [C] = \text{J K}^{-1} \qquad [c] = \text{J kg}^{-1} \text{K}^{-1}$$

Material	$c / \text{J kg}^{-1} \text{K}^{-1}$
copper	385
water	4186
aluminium	900
iron	450

Write the balance and solve for the equilibrium temperature.

$$m_1 c_1 (T_1 - T_{eq}) = m_2 c_2 (T_{eq} - T_2) \quad \implies \quad T_{eq} = \underline{\hspace{5cm}}$$

Using $m_1 = 0.2 \text{ kg}$ copper at 80 °C and $m_2 = 1.0 \text{ kg}$ water at 20 °C, calculate T_{eq} .

Is the result closer to 80 °C, 50 °C, or 20 °C? Explain using both mass and specific heat capacity, not just one of them.

5. Conservation isn't enough

a thought experiment

Imagine a Tuesday on which the copper ball spontaneously grows hotter while the water cools further, energy flowing the “wrong” way. Every joule leaving the water arrives in the copper.

Has conservation of energy been broken? _____

If not, what has been broken?

6. Counting returns

energy quanta shared between two objects

Model each object as independent oscillators, each able to hold any whole number of identical energy quanta. Object A has 2 oscillators; object B has 4. The number of ways to share q quanta among n oscillators is

$$\Omega(q, n) = \binom{q + n - 1}{n - 1}.$$

Complete the table for a total of $Q = 6$ quanta, $q_A + q_B = 6$.

q_A	q_B	$\Omega_A(q_A, 2)$	$\Omega_B(q_B, 4)$	product
0	6			
1	5			
2	4			
3	3			
4	2			
5	1			
6	0			

Which split is most numerous? _____ Ring it in the table.

7. The majority verdict

three questions

1. Did anything push the quanta toward the dominant split?
2. Was energy conserved throughout every split you tried?
3. So why is that split special?

8. Where temperature comes from

extension

At the maximum, moving a small energy dE from A to B changes the total count by nothing.

$$d\Omega_{\text{total}} = 0 \quad \Rightarrow \quad \frac{d(\ln \Omega_A)}{dE_A} = \frac{d(\ln \Omega_B)}{dE_B}$$

Using $S = k_B \ln \Omega$, rewrite this as a statement about entropy.

Thermodynamics defines $1/T \equiv dS/dE$. Complete the chain.

microstates $\Omega \rightarrow$ entropy $S = k_B \ln \Omega \rightarrow$ temperature _____

9. The reframing

observed and inferred

Fill in what the apparatus actually shows versus what is being inferred about it.

Observed

Inferred

In one sentence, explain why “the system ran out of things to do” is the wrong way to describe equilibrium.

10. Core results

what this enquiry actually earned

The relations established along the way.

$$C = mc \qquad T_{eq} = \frac{m_1 c_1 T_1 + m_2 c_2 T_2}{m_1 c_1 + m_2 c_2} \qquad \Omega_{\text{total}}(q_A) = \Omega_A(q_A) \Omega_B(q_B) \qquad \frac{1}{T} \equiv \frac{dS}{dE}$$

Concepts earned, in order.

- Equal temperature is how equilibrium is recognised from outside, not what it is.
- Thermal capacity splits into an extensive ingredient, mass, and an intensive one, specific heat capacity.
- Conservation of energy alone does not forbid energy flowing the “wrong” way; something else does.
- Equilibrium is the overwhelmingly numerous split of energy, found by counting, not by anything pushing.
- Equal temperature is a consequence of maximising Ω , not an assumption fed into the argument.

Every substantial claim should be accompanied by a reason, calculation, table, or counterexample.